

Research article

Ameliorating effects of berberine on MK-801-induced cognitive and motor impairments in a neonatal rat model of schizophrenia

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ABSTRACT

Neonatal administration of MK-801 (NMDA receptor antagonist) results in schizophrenia-like behaviors in rodents. Berberine (BBR) is a herbal alkaloid, which shows many neuroprotective properties in neurodegenerative diseases. The present study was designed to clarify whether systemic administration of BBR improves motor and cognitive disturbances induced by MK-801 treatment. Male Wistar rat pups were treated with intraperitoneal administration of saline (1 ml/kg) as a control group, MK-801 (1 mg/kg), BBR (20 mg/kg) and BBR (20 mg/kg) plus MK-801 (1 mg/kg). Treatments were administered on postnatal day (P) 6–10 for once daily. To assess motor learning, coordination as well as spatial learning and memory, behavioral evaluation was performed at P55–60, using the rotarod, open field, and Morris water maze paradigm. MK-801 injection led to motor perturbations in both the open field and accelerating rotarod tests, which were restored by BBR. Also, BBR improved learning impairments, although it had no significant effect on the Probe test. Taken together, it can be concluded that BBR produces a neuroprotective effect in rats with MK-801-associated behavioral deficits. Given that the MK-801 exposure demonstrates an animal model of schizophrenia, we suggest that timely BBR administration may act as a potential treatment in schizophrenic patients.

1. Introduction

Berberine chloride hydrate (BBR), a plant-derived alkaloid, has multiple pharmacological properties including neuroprotective, antioxidant, anti-apoptotic and anti-inflammatory effects, as well as brain blood flow improvement [1]. Evidence suggests that BBR reduces neuronal insult in rodent models of Parkinson's disease, brain ischemia, stroke, and encephalomyelitis [2–5]. Previous studies have attempted to show that BBR administration decreases cell death and neural differentiation in the hippocampus, a key brain center for learning and memory [6]. BBR has also been reported to stimulate the survival of neuronal cells expressing *N*-methyl-[D]-aspartate (NMDA) receptors [7]. In a cellular and animal study conducted by Lee et al. (2010), it was shown that BBR elicited neuroprotective properties in MK-801-exposed rat pups [8].

MK-801 (dizocilpine maleate), a potent and noncompetitive antagonist of the NMDA receptor, exerts both neuroprotective and neurotoxic activities, which may be due to various conditions such as the

route, dose, and schedule of administration [9]. It has been revealed that MK-801 induces a behavioral syndrome including hyperlocomotion, cognitive deficits, stereotyped movements, and ataxia, which represents certain aspects of schizophrenia in humans [10]. Also, animal investigations have shown that neurotrophic factors and neurotransmitter systems such as glutamate, dopamine, acetylcholine, and GABA may be altered by MK-801 administration, similar to that of people with schizophrenia [11].

Schizophrenia as a debilitating mental disorder seems to affect approximately 21 million people worldwide (WHO, 2018). This disorder is known to involve different enzymes and neurotransmitter systems such as prolyl endopeptidase (PEP) and glutamate, respectively [12,13]. PEP is a cytosolic peptidase that extensively dispensed in different brain regions and other organs. Previous studies have established alterations of PEP in psychiatric disturbances such as schizophrenia and depression [14]. For instance, Frenssen et al. reported a negative correlation between the serum PEP level and severity of depression, whereas the intensity of signs in schizophrenic patients was directly

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related to the higher plasma activity of PEP [15]. MK-801 administration as a schizophrenia-relevant behavioral phenotype inducer in rodents affects PEP activity in different brain regions. In an investigation performed by Ahmed et al., it was shown that a high dose of MK-801 significantly increased the activity of PEP in several brain areas such as the frontal cortex [9]. More recently, PEP inhibitors have gained importance as a significant target for the treatment of mental diseases, and therefore may have clinical implications. Surveys such as that conducted by Tarrago et al. have shown that berberine can directly inhibit PEP in a dose-dependent manner, suggesting new perspectives for the treatment of neuropsychiatric diseases [16].

Based on (1) the evidence which shows that BBR may modulate the function of the glutamate system; and also (2) controversial effects of BBR and MK-801 on neuroprotection and PEP level; as well as (3) considering that current treatment options for all symptoms of schizophrenia are not effective enough, the present study set out to assess the effects of berberine on spatial learning and memory, motor coordination and locomotor activity in the postnatal MK-801 model of schizophrenia in rats.

2. Materials and methods

2.1. Animals

Wistar rats were kept in groups of 4 per cage in a temperature-controlled environmental condition ($21 \pm 2^\circ\text{C}$) on a 12/12 h light/dark cycle. They were given free access to standard food and water, except during the time of the experiments. A total of eight or ten animals were used in each group, and each rat was analyzed only once. All experiments were performed between 9:00 AM and 13:00 PM. Maximum efforts were made to lessen the number of rats and their suffering. The animal care and behavioral experiments were conducted according to the “NIH Guide for the Care and Use of Laboratory Animals” and approved by the ethical committee of Kerman University of Medical Sciences, Iran (approval number: IR.KMU.REC.1395.819).

2.2. Experimental procedure

The experiments were performed on forty male rat pups at 55 days of age weighing 100–120 g from 10 pregnant female Wistar rats. Male pups divided randomly into four groups: control (saline-treated 1 ml/kg), BBR (20 mg/kg) [17], MK-801 (1 mg/kg) [18], and MK-801 + BBR received MK-801 (1 mg/kg) plus BBR (20 mg/kg) with a one hour interval. It should be considered that all treatments were given intraperitoneally on postnatal day (P) 6–10 once a day [18], and all behavioral experiments were carried out at P55–60. BBR and MK-801 were purchased from Sigma-Aldrich and Tocris, respectively. All drug solutions were dissolved in 0.9% saline.

2.3. Rotarod test

The rotarod, which is composed of an accelerating rotating rod, is an accepted apparatus for evaluating behavioral parameters indicating motor coordination and balance in rats (Hugo Sachs Elektronik, Germany) [19]. The experiment began from a minimum speed of 10 revolutions per minute (rpm) to the maximum speed of 60 rpm. Animals were placed on the rod, and three 5 min trials, with a 30 min interval between trials were carried out for each rat. Latency to fall as the standard index of balance and motor coordination was recorded [20].

2.4. Open field test

The apparatus was basically the same as described previously [20]. An open field apparatus was made of plexiglass and consisted of a square arena ($90 \times 90 \times 30$ cm), and its floor was divided by lines into 16 equal squares. All experiments were performed in a dimly

illuminated testing room. Each rat was individually placed in the central zone and allowed 5 min of free exploration. All sessions were recorded and then analyzed using Ethovision software (version 7), a video tracking system for automation of behavioral tests (Noldus Information Technology). The following behavioral indices were then evaluated: total distance moved (cm); total duration of mobility (s) and immobility (s); the number of rearing (as a measure of vertical activity). The apparatus was cleaned after testing each rat [21].

2.5. Morris water maze test

The Morris water maze is a valid apparatus to evaluate spatial learning and memory in rodents [22]. This apparatus is a large circular tank, 140 cm in diameter and 45 cm in depth, filled with water ($22\text{--}24^\circ\text{C}$). The animals ran away from water onto a hidden platform (15 cm wide, 35 cm high), which placed 1.5 cm beneath the water level. The water maze was surrounded by various visual cues on the wall of the testing room, and their location remained unaltered all over the test period. The apparatus was divided into four quadrants, and the rats were randomly placed in one of the four equal quadrants. The animal's behavior in each experiment was video-recorded by a camera which was placed above the center of the water pool. The spatial learning and memory-related parameters such as the total time spent to detect the hidden platform (escape latency) and the distance moved to reach the hidden platform (traveled distance) were measured by a video-tracking system software (Ethovision, Noldus Information Technology, Netherlands) [23].

The training session included three blocks with a 30 min interval and each block comprised of four consecutive trials. On each trial, animals were randomly dropped into the tank from a determined point of each quadrant. The animals were allowed to swim for 60 s to detect the hidden platform. After the exploration of the platform, the rat remained there for 20–30 s and then was caged for 20–30 s before the next trial.

The retention of spatial memory was evaluated two hours after training trials by removing the platform in a 60 s probe trial. Total time spent and distance traveled in the target quadrant which previously contained the platform were recorded as the indices of spatial memory retention [20,23].

2.6. Statistical analysis

Normality and homogeneity of variances were checked using Shapiro-Wilk and Leven tests, respectively. A repeated-measures ANOVA and Bonferroni post-hoc test were used to compare motor learning in rotarod and learning phase in the Morris water maze. All other obtained data were analyzed using one-way ANOVA followed by Tukey's post-hoc analysis test. Results were reported as mean $\pm$ SEM. In all statistical comparisons, p values less than 0.05 was considered as the criterion for statistical significance. All data analyses were performed using the SPSS software package version 22.0 (SPSS Inc, Chicago, IL, USA).

3. Results

3.1. The effects of BBR on motor learning and coordination of rats exposed to MK-801 in the rotarod test

Mean latency of three trials on accelerating rotarod has been shown in Fig. 1a. One-way ANOVA analysis indicated that different groups of rats showed a significant difference in the mean performance [$F(3, 35) = 6.38$, $p < 0.01$, Fig. 1a]. Post hoc comparisons showed that intraperitoneal administration of MK-801 significantly decreased time on the rod, indicating motor symptoms ($P < 0.05$, MK-801 compare to control). The results revealed no significant difference between BBR + MK-801 compared to control groups and also between BBR +

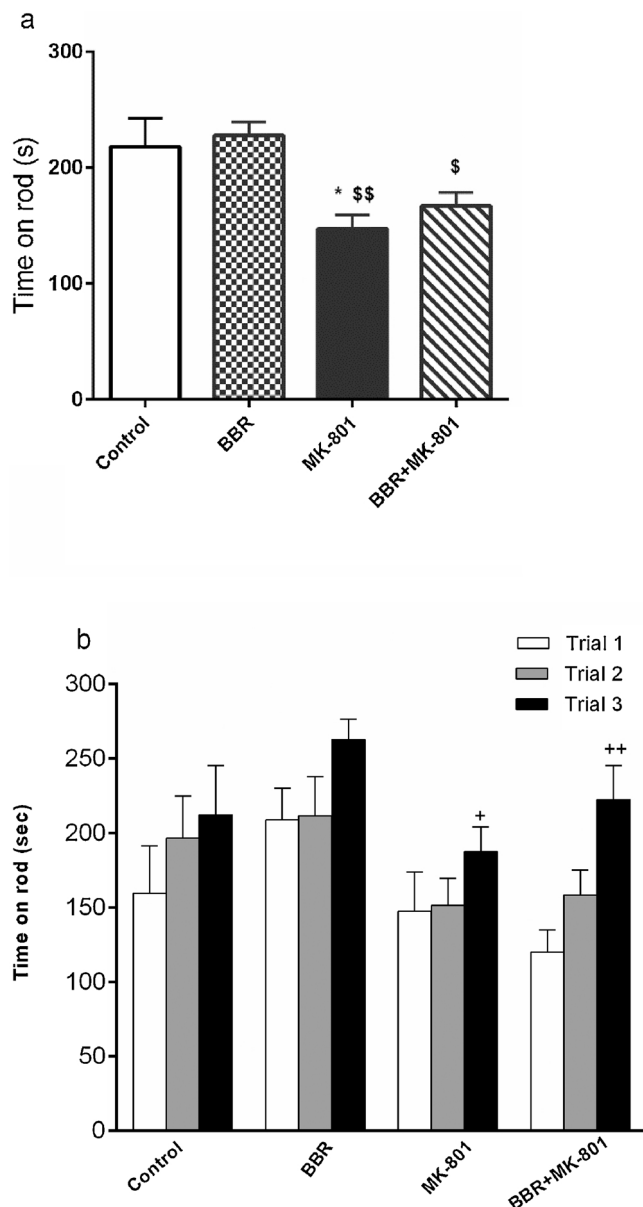


Fig. 1. The effects of intraperitoneal injection of berberine (BBR) on MK-801-induced deficits in the rotarod test. Mean latency of three trials of staying on the rod (a), Falling time in three successive trials (b). Significant differences: * $p < 0.05$ compared to control group. \$ $p < 0.05$ and \$\$ $p < 0.01$ compared to BBR group. + $p < 0.05$ and ++ $p < 0.01$ compared to 1st trial. Each bar represents mean $\pm$ SEM ($n = 8$ in control and BBR groups; $n = 10$ in MK-801 and MK-801 + BBR groups).

MK-801 compared to MK-801 rats ($p > 0.05$). BBR alone caused no significant change in the maximum motor performance of rats in rotarod task ($p > 0.05$).

Fig. 1b illustrates the task acquisition in different groups. A significant improvement in the task acquisition was observed between trial 3 and trial 1 for MK-801 and BBR + MK-801 rats [$F(2, 64) = 14.01$, $p < 0.01$ for EE + MK-801 and $p < 0.05$ for MK-801 groups], but not for BBR and control groups. This discrepant effects may be due to the better performance at the first trial for BBR and control rats.

3.2. The effects of BBR on the locomotor and exploratory activity of rats exposed to MK-801 in the open field test

Fig. 2 provides the results obtained from the locomotor activity of animals in the open field test, as assessed by the total distance moved and velocity. One-way ANOVA indicated that there was a significant difference among groups of study ($F(3,39) = 17.70$, $p < 0.01$). As shown in Fig. 2a, postnatal administration of MK-801 significantly increased the total distance moved in comparison to the control group ($p < 0.01$), suggesting a hyperactive effect of MK-801 in the rats in the open field test. BBR by itself decreased this parameter ($p < 0.05$). Furthermore, BBR improved hyperactivity associated with MK-801 exposure (BBR + MK-801 rats vs. MK-801, $p < 0.001$), and even led to the hypoactivity in this group (BBR + MK-801 rats vs. control rats, $p < 0.05$).

Fig. 2b illustrates the effects of BBR alone or in the presence of MK-801 on velocity in the open field test. One-way ANOVA revealed that there was a significant difference among experimental groups ($F(3,39) = 14.41$, $p < 0.001$). Postnatal administration of MK-801 significantly increased velocity in comparison to the control group ($p < 0.01$), whereas BBR in combination with MK-801 decreased this parameter ($p < 0.05$). Furthermore, intraperitoneal administration of BBR by itself or in the presence of MK-801 decreased velocity in comparison to the MK-801 group ($p < 0.001$).

Fig. 2c shows the exploratory behavior of rats which was measured by vertical locomotor activity (number of rearing and climbing). One-way ANOVA analysis indicated that the rearing of rats between different groups was significant [$F(3,39) = 5.301$, $p < 0.01$]. What stands out in the figure is a significant increase in rearing number in BBR ($p < 0.01$) MK-801 ($p < 0.05$) and BBR + MK-801 groups ($p < 0.05$) in comparison to the control group.

No significant difference for the duration of time spent in the central and peripheral area and also the duration of mobility and immobility between groups was evident ($p > 0.05$) (Data not shown).

3.3. The effects of BBR on the learning and memory of rats exposed to MK-801 in the Morris water maze

The effects of intraperitoneal injection of BBR alone or in the presence of MK-801 on spatial learning in the Morris water maze are presented in Fig. 3. During the training phase, the animals of all groups found the hidden-platform faster in the 3rd block than in the 1st block. A remarkable reduction in the traveled distance ($p < 0.001$ for all groups, Fig. 3a) and time ($p < 0.001$ for control & MK-801 + BBR groups, $p < 0.01$ for BBR and $p < 0.05$ for MK-80, Fig. 3b) presents progress in the acquisition of learning task.

Comparison among groups in the 3rd block showed a significant difference among groups ($F(3,36) = 5.3$, $p < 0.05$). Post hoc analysis revealed a significant increase in both time and distance traveled to detect the submerged platform in MK-801 animals in comparison to the control group ($p < 0.01$, Fig. 3a & b), proposing a major defect of spatial learning for MK-801-treated rats. None of these differences were found between BBR + MK-801 and control group ($p > 0.05$, Fig. 3a & b). No significant reduction in the time and the distance traveled in the BBR group compared with control rats was evident ($p > 0.05$, Fig. 3a & b). Rats in BBR + MK-801 group moved significantly shorter distances and spent less time to detect the platform compared to MK-801 rats, indicating spatial learning improvement for combination group ($p < 0.05$, Fig. 3a).

As can be seen from Fig. 3a & b in the 1st and 2nd blocks, no significant differences in distance and time to find the platform between groups was detected. It should be noted that statistics for comparison among blocks has not been shown on the graph.

In the recall probe, memory performance was evaluated using exploring the target quadrant in which the platform was situated during the training phase. Results, as measured by time and percentage of

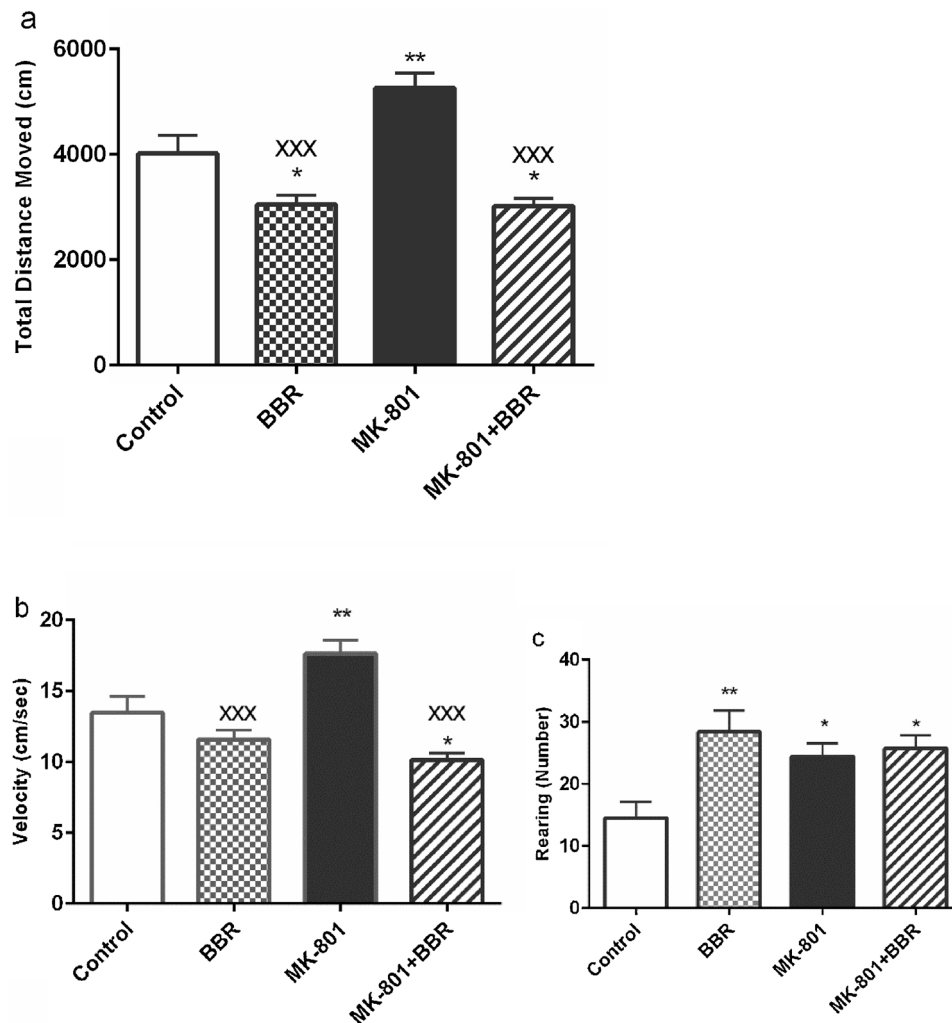


Fig. 2. The effects of intraperitoneal injection of berberine (BBR) on the locomotor and exploratory activity of rats exposed to MK-801 in the open field test. Total distance moved (a), velocity (b), rearing (c). Significant differences: * $p < 0.05$ and ** $p < 0.01$ compared to control group. $\times \times \times p < 0.001$ compared to MK-801 group. Each bar represents mean $\pm$ SEM ($n = 10$ in each group).

traveled distance in the correct quadrant, indicated that MK-801-treated rats remembered the platform location significantly less than that of control rats ($p < 0.05$ for both time and distance, Fig. 3c & d). These measures, which show memory retention, has not been significantly different between BBR + MK-801 and MK-801 group ($p > 0.05$, Fig. 3c & d). No significant difference between groups for crossing number to the correct quadrant was evident ($p > 0.05$). Swimming speed was recorded throughout the experiment, and one-way ANOVA did not show any significant difference in the mean swimming speed among the groups, suggesting no influence of swimming speed on the latencies ($p > 0.05$) (Data not shown).

4. Discussion

Our results demonstrated that berberine (BBR) treatment, as a herbal medicine, modified balance perturbation, locomotor agitation and spatial learning disturbances induced by neonatal MK-801 administration in a rat model of schizophrenia, suggesting a beneficial effect of timely BBR exposure in MK-801-associated cognitive deficits. The current study confirmed those findings that demonstrate the first three weeks after birth, especially postnatal days 7–14, are considered as a critical period for brain development and NMDA receptor expression in rodents [24]. Furthermore, previous studies indicated that exposure to NMDA antagonists such as MK-801 during this period might simulate

some core symptoms and etiological aspects of schizophrenia [25]. NMDA receptor (NMDAR) signaling is one of the main pathways in cognition, locomotion, synaptic plasticity, learning and memory [26–28], and its dysfunction in different brain regions, especially in the prefrontal cortex (PFC), represents schizophrenia-like behavior [29]. NMDA receptor antagonists have been extensively used to model schizophrenia in laboratory rodents [30]. Chronic systemic administration of NMDAR antagonists during the first two weeks after birth can lead to defect in sensorimotor gating, cognitive function, and locomotor activity at adulthood [31]. A large body of investigations suggests that the chronic use of these antagonists alters some neurotransmitter systems involved in cognitive behavior including GABAergic and dopaminergic systems [32–35].

The findings of the rotarod test indicated that systemic BBR administration modified MK-801-induced balance disturbance in rats. As formerly demonstrated, this test is a valuable paradigm for the interpretation of motor skill learning in rodents [19]. Using a rat model of tremor, Vaziri et al. (2015) showed that intraperitoneal injection of BBR (20 mg/kg) partially ameliorated harmaline-induced imbalance in the rotarod test. It has been suggested by authors that this improvement in the balance disturbance may be mediated through the blocking of NMDA receptors [17]. In the extant study, this research hypothesis was tested by administering MK-801 as an NMDAR antagonist. According to our achieved results, it seems that the effect of BBR on locomotor

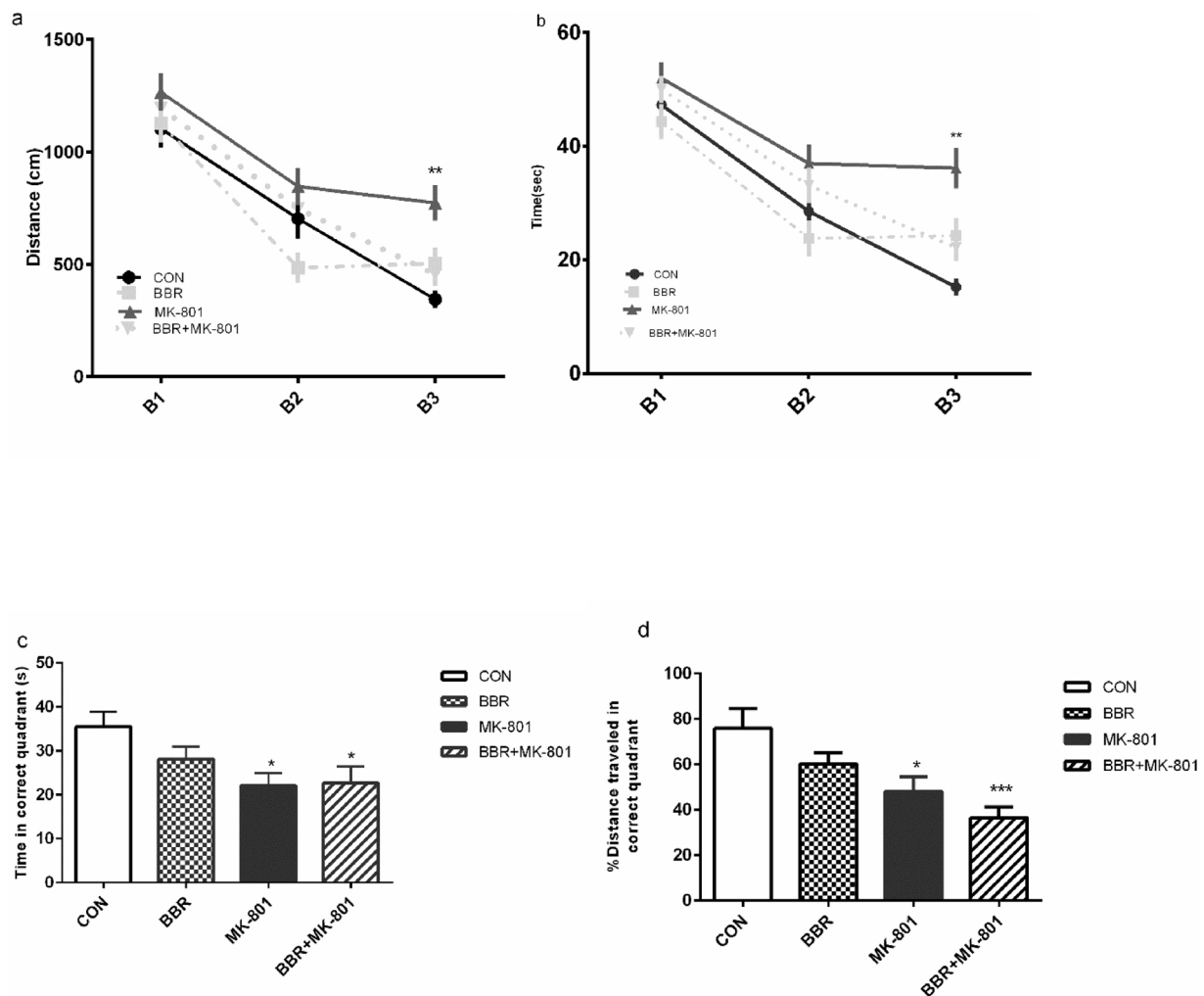


Fig. 3. The effects of intraperitoneal injection of berberine (BBR) on spatial learning of rats exposed to MK-801 in the Morris water maze task. Total distance moved in each block (a), total time spent in each block (b), Time spent in the target quadrant (c), distance traveled in the target quadrant (d): Significant differences * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ compared to control group. Each bar represents mean $\pm$ SEM ($n = 9$ in control, BBR, and MK-801 groups; $n = 10$ in MK-801 + BBR group).

coordination is not related to the NMDA receptors. This discrepancy can be attributed to the involvement of other neurotransmitters such as GABA and dopamine. Although MK-801 increases dopamine, previous studies show that imbalance at the rotarod test is known as a “non-dopamine dependent” motor function [36,37]. Considering that various receptors and neurotransmitters may be involved in the effects of BBR on motor coordination in the rotarod test, elucidating the exact mechanism underlying this effect will require further studies.

In the open field test, MK-801 injection showed a marked increase in the distance moved, velocity and rearing with no significant effect on the time spent in the center and other aspects of the locomotor activity. BBR alone or in combination with MK-801 significantly decreased the total distance moved and velocity. Our data may be in line with a report that suggests intraperitoneal administration of BBR (10, 20 and 50 mg/kg) dose-dependently reduces parameters as mentioned above and has a notable effect on motor function [17]. In contrast, in a rat model of streptozotocin-induced Alzheimer's-like dementia, Oliveira et al. reported that oral gavage of BBR (50 or 100 mg/kg) once daily for 21 days made no significant change in locomotor and exploratory activities using the open field test [38]. It should be noted that the inconsistent responses of BBR on locomotor activity may be due to various doses, route and time courses of administration and also different experimental models in animal species.

In the current study, the Morris water maze test as a reliable

paradigm of cognitive function was used to assess the potential protective effect of BBR on spatial learning and memory in the MK-801 model of schizophrenia. An interesting finding was that BBR prevented the learning impairments following postnatal MK-801 treatment, but had no effect on probe trial performance. In accordance with the present results, several studies have demonstrated that systemic administration of BBR can ameliorate the learning and memory deficits in different experimental models of cognitive impairments in rodents by using a Morris water maze test task [39–41]. For instance, it has been reported that oral BBR administration at a dose of 50 mg/kg improved learning and memory deterioration due to the heavy metals-induced Alzheimer's-like disease in rats [42]. Also, several investigations showed that BBR decreases latency time and modifies learning and memory dysfunction via improving the oxidative stress and inflammatory factors [42,43]. It is therefore likely that learning improvement by BBR in MK-801 model of schizophrenia at least partly may be mediated through anti-oxidant and anti-inflammatory activity of this plant alkaloid. Further work is required to establish the viability of this issue.

A large number of studies have supported the remedial potential of BBR in different brain diseases including brain ischemia and stroke [5,41,44], Parkinson's [45], and Alzheimer's disease [5,46]. The neuroprotective effects of BBR may be ascribed to its role in the induction of neurogenesis or the excitatory/inhibitory (E/I) circuit balance during

the critical period of development [8,9]. Previous studies showed that berberine could decrease the action potential threshold via blockade of potassium channels, depolarize synapses and activation of NMDA receptors [8,9]. Despite the protective effects of BBR such as the antioxidant, neuroprotection, neurotransmitter balance and also learning and memory improvement, it tends to accumulate in the nervous system at a high dose and produce neurotoxicity [47]. Considering that there are some controversial issues in the literature about the safety of BBR, consecutive cellular and histological studies are suggested for the future.

5. Conclusion

In summary, the combination of our findings provides some support for the beneficial effects of BRB on preventing cognitive and motor disturbances associated with the early postnatal administration of NMDA blocker. Therefore, BRB may act as a potential clinical agent, which can protect and ameliorate some behavioral impairments in schizophrenia. Though we did not evaluate the possible signaling pathways involved, the precise mechanism of the neuroprotective effect of BBR in MK-801 model of schizophrenia remains to be elucidated.

Conflict of interest

The authors declare no conflict of interest.

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